

Technical Specification: Multilingual Embedding Engine

1. System Alignment (FR-3)

Fulfills Functional Requirement 3 (Embedding Generation). The component abstractly decouples text processing from database transaction execution, handling mathematical coordinate mapping for both document parsing and runtime search operations.

2. Model Specification

- Model Core identifier: paraphrase-multilingual-MiniLM-L12-v2
- Underlying Architecture: Multilingual MiniLM Transformer network
- Vector Output Dimensionality: 384 Dimensions

3. Architectural Enhancements for Localization

While standard systems often deploy English-centric architectures (such as all-MiniLM-L6-v2), this project upgrades to a multilingual model framework. This provides native support for over 50 regional languages, allowing the model to handle corporate governance and structural documentation written in English, Hindi, and Malayalam.

4. Vector Database Mapping

The model's 384-dimensional floating-point array maps exactly to the custom pgvector data type defined in the database schema layer: embedding vector(384). Prior to data delivery, numpy matrices are flattened into standard Python native lists using the .tolist() method, ensuring compatibility with the database driver.